Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

David Afonso Valente
Bottlecap AI

Abstract

Base models are usually chosen by public benchmark scores. Vendors optimise for those scores, their items can leak into training data, and they grade a model before the fine-tuning it will receive. We show that a short, unlabelled adaptation to the target domain predicts how models will rank after fine-tuning, before any task data exist. Every candidate is adapted the same way to fresh text from the domain and ranked by how well it then predicts held-out text. This takes minutes to an hour per model on one GPU. Ranked this way, fifteen base models land within one rank of where they finish once fine-tuned on a news-writing task. We also show that adaptation reorders the models, lifting some that look weak before adaptation to the top, and that the adapted ranking, with no fine-tuning, agrees with the benchmarks built for the domain. Adapted on general web text and news, it converges on HellaSwag, the likelihood-scored benchmark closest to language modelling; adapted on arXiv mathematics, it moves towards the reasoning benchmarks GSM8K and MMLU-Pro. The ranking is stable across adaptation seeds and hardware and requires no labelled data. Code and artifacts are released.

1 Introduction

Suppose an engineer has to pick a base model for a product. They have plenty of text from the product's domain and a shortlist of a dozen open checkpoints, and the model they pick will be fine-tuned and shipped. The most reliable way to choose would be to fine-tune every candidate and compare the results, but that needs a task definition, labelled data and a training run for each model. In practice the choice is made from benchmark scores instead: MMLU-Pro, a set of knowledge questions; HellaSwag, which asks for the most natural ending of a passage and is scored by likelihood; and GSM8K, grade-school maths. In this paper we show that the fine-tuning ranking can be predicted before any task exists, from unlabelled text alone. We adapt every candidate in the same way to text from the domain and then rank the candidates by how well they predict held-out text from it. Since the text can be collected fresh for each domain, we call this a *dynamic evaluation*. The resulting ranking matches the ranking after fine-tuning almost exactly. Of the benchmarks, only HellaSwag comes close.

Benchmark scores are misleading for two reasons. The first is that benchmarks are public. They were written for other purposes and released before the models they now evaluate, vendors optimise for the ones they report, and their questions can end up in pre-training data. A high score may come from a stronger model or from a model that has seen more of the test, and because pre-training data is rarely disclosed, a reader cannot tell which. Benchmarks can also disagree sharply with how well a model handles ordinary text. Llama-3.2-1B and Qwen-2.5-1.5B are 2.1 points apart on HellaSwag and their loss on news text after adaptation differs by only 3.4%, but they are 55.3 points apart on GSM8K (accuracy, in percentage points).

The second reason is that models are rarely used as they ship. Teams adapt a base model to their own data first, so what matters is how good it becomes after adaptation. A model that has seen little text like the target domain can look weak at first and still end up ahead. Gemma-4-12B is an example: among the fifteen

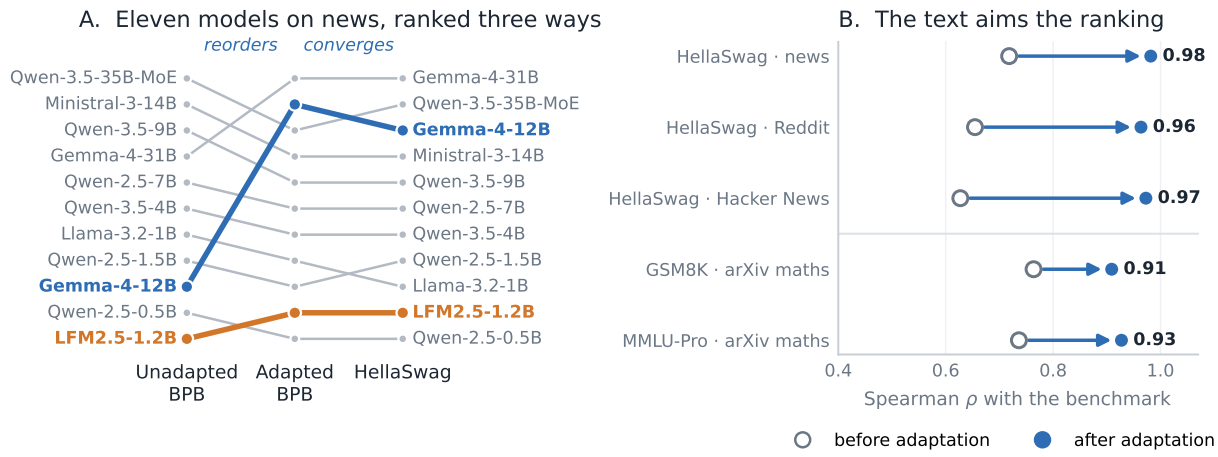


Figure 1: Adaptation reorders the models, and the text determines what the ranking follows. **A**: the eleven models evaluated on all four corpora, ranked by news bits per byte (BPB; lower is better) before adaptation, after adaptation, and by HellaSwag. Many lines cross between the first two columns, as Gemma-4-12B moves from ninth to second and LFM2.5-1.2B leaves last place. Between the last two columns the lines are almost parallel, so the adapted order is close to HellaSwag's. **B**: rank agreement (Spearman ρ; 1 means lower BPB goes with a higher score) between BPB and each benchmark, before adaptation (hollow) and after (filled). After adaptation on general text the ranking agrees with HellaSwag on all three corpora; after adaptation on arXiv mathematics it agrees with GSM8K and MMLU-Pro. The full matrix is Table 4 in Appendix B.

fine-tuned models it ranks ninth on GSM8K, eighth on MMLU-Pro and thirteenth on loss before adaptation, second on loss after adaptation, and it produces the best of the fifteen fine-tuned systems.

We argue that loss is the right measure. There is no universal ground truth for which base model is better: a benchmark tells us which model is better at that benchmark, and different benchmarks disagree. A base model learns what it can do by minimising cross-entropy on text, and adaptation minimises the same loss on the domain. Held-out loss is therefore the objective each candidate was last trained on, measured on text it was not trained on. Cross-entropy is also a proper scoring rule. If p is the distribution that generated the domain's text and q is a model, the model's expected loss is

$$H(p,q) = H(p) + \mathrm{KL}(p \,\|\, q), \tag{1}$$

where $H(p)$, the entropy of the text, is the same for every model, and $\mathrm{KL}(p \,\|\, q)$ measures how far the model's predictions are from the text's distribution. The difference between two models' expected losses is therefore the difference in their KL divergence from the text, and held-out loss estimates it [1].

Three design choices keep the comparison fair. We count the loss per byte instead of per token, because each model has its own tokenizer but they all read the same bytes; we call the result *bits per byte* (BPB). We compute it on fresh, decontaminated text that none of the candidates has trained on. And we give every candidate the same adaptation procedure on the same text, so that each model is evaluated in the state it would be used in. Train-before-test [16] also prepares all candidates identically before comparing them, but it uses labelled benchmark data. Our approach needs no labels, uses fresh text and measures in bytes.

We evaluate seventeen base models from eight families, from 0.5B to 35B parameters, on four corpora: news, Reddit, Hacker News and arXiv mathematics. Fifteen of the models are also fine-tuned on a news task, writing an article's opening paragraph from its body. Our main findings are:

- **Adapting to the domain predicts fine-tuning on the task.** When the models are ranked by adapted BPB before any fine-tuning, every model ends up within one rank of its position after fine-tuning. HellaSwag comes close and the reasoning benchmarks do not. Adapting on Reddit instead of news predicts the news

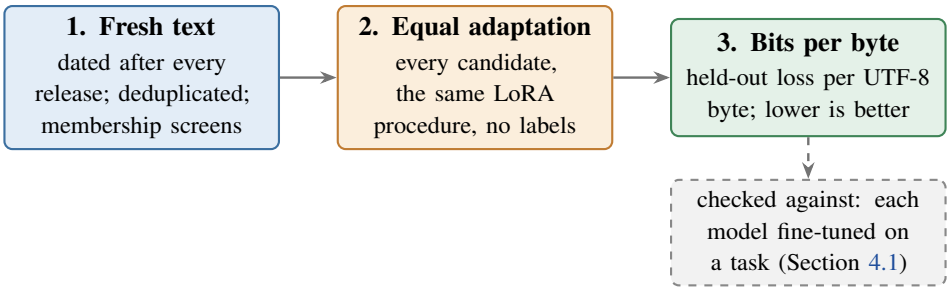


Figure 2: The method. The corpus is fixed and screened first; every candidate is then adapted in the same way and scored on held-out text. The fine-tuning experiment of Section 4.1 is not part of the method. We use it only to check the method's predictions.

fine-tune equally well, so the prediction does not depend on adapting to the task's own text (Figure 3, Section 4.1).

- **Adaptation reorders the models.** Loss before adaptation partly reflects how familiar a model already is with the style of the text. Adaptation removes most of that effect, and some models that look weak beforehand move to the top (Figure 1A, Section 4.2).
- **The choice of text determines what the ranking measures.** On general text the adapted ranking agrees closely with HellaSwag; on mathematics it agrees with the reasoning benchmarks GSM8K and MMLU-Pro (Figure 1B, Section 4.3).
- **The ranking is stable.** It barely changes across adaptation seeds, hardware and general-text corpora, while benchmark scores for the same weights shift when only the evaluation machine changes (Section 4.4).

All numbers in the paper can be recomputed from the released artifacts at

`https://github.com/DavidAfonsoValente/entropy-bench`

2 Method

The method has three phases (Figure 2). The corpus is collected and screened before any model is run on it, then every candidate is adapted in the same way, and finally each adapted model is scored on held-out text.

Phase 1: a decontaminated corpus. The evaluation text must be text that no candidate has seen, and we use three safeguards for this. *Time:* the text is dated after the candidates' release. Reddit and Hacker News were scraped in August 2026, after every model in the cohort was released, and the arXiv titles and abstracts run from 9 June to 20 August. *Hygiene:* we remove exact duplicates, text that fails the Gopher quality heuristics [12] and MinHash/LSH near-duplicates at Jaccard 0.60. The 80/10/10 split shares no identifier or normalised text across splits, so nothing in the test split reappears in training. *Membership screens:* the news snapshot (Google News, 8 June 2026, 119,054 articles) predates some of the releases, so we also test every model for having seen it. Three reference-free tests look for traces of memorisation. Min-K++ [10] flags passages whose least likely tokens are unusually likely for the model. A context-sensitivity test flags passages whose loss drops more than expected when the model is shown other text from the same corpus. A perturbation test checks whether the model prefers the original passage over copies with swapped words. We remove every passage flagged for any model, 80 in total, so all candidates are measured on the same corpus (Appendix C).

Phase 2: equal adaptation. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, using LoRA [28] on all linear layers. In the full method each candidate gets its own hyperparameter search with Optuna [25]. A Tree-structured Parzen Estimator proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never on the test split. Every model gets the same number of trials and no time limit. With a wall-clock limit, small models complete more trials than large ones and end up better tuned, a bias we observed in an earlier run (Appendix C). The best configuration is then trained until validation BPB stops improving. The released pipeline implements this procedure (`--hparams sweep --no-time-limit`).

Phase 3: bits per byte. Cross-entropy on the held-out test split is divided by the UTF-8 byte length $|B|$ of the text those predictions cover:

$$\mathrm{BPB} = \frac{\mathscr{L}_{\mathrm{nats}}}{\ln 2} \times \frac{|\mathrm{Tokens}|}{|B|} \tag{2}$$

The byte count depends only on the text, so it is the same for every candidate, and a larger vocabulary gives no advantage as it would with per-token perplexity. Test blocks do not overlap, and each starts with no earlier context. Beginning-of-document tokens between packed documents are used as context but never scored. We report BPB both before adaptation, which we call unadapted (or zero-shot) BPB, and after adaptation, since the difference between the two is informative (Section 4.2).

3 Experimental Setup

Models. We evaluate 11 base checkpoints between 0.5B and 35B parameters on all four corpora (Table 9). Most are dense Transformers; one is a sparse mixture-of-experts and one, LiquidAI's LFM2.5-1.2B, is not a Transformer. For the news corpus we add six more models, chosen in advance to fill gaps in model size and to add publishers, for 17 in total (Table 8). They come from the Llama, Qwen, Gemma, Ministral, LiquidAI, Falcon, Mistral and SmolLM families [29, 30, 31, 32, 33].

Corpora. We use four corpora: news, Reddit, Hacker News and arXiv mathematics. Table 3 gives their sizes and how each was screened.

Adaptation runs. Because of limited compute, all runs reported here use a single hand-picked configuration instead of the per-model search: LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch size 32 and 250 update steps. This is 4.1M training tokens per candidate and costs 3.7 GPU-minutes on one A100 at 0.5B parameters, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B. The configuration is close to what our earlier search found (rank 16 won for eight of the eleven models, and the median winning learning rate was 1.1×10^{-4}), and its results are stable across seeds and hardware (Section 4.4). Adaptation ran on Leonardo (CINECA), on 3 nodes with 4 A100 64 GB GPUs each.

Benchmarks. For comparison, we score every model on MMLU-Pro [19], HellaSwag [20] and GSM8K [21] with *lm-evaluation-harness* 0.4.12 [13], on one machine with one software build, following the protocol in Appendix F. We use scores we measured ourselves, not the values reported on model cards, and we publish the harness logs so every score can be recomputed.

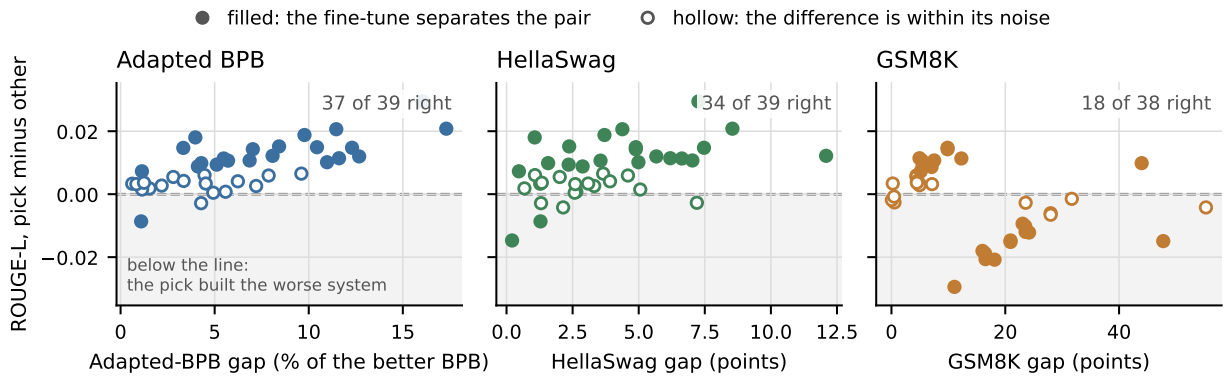


Figure 3: How well does each ranking predict the fine-tuned results? Each point is a pair of fine-tuned models whose parameter counts are within a factor of two (39 pairs; 38 for GSM8K, where two models tie). For each pair, x is how strongly the ranking prefers one model (the gap in that measure) and y is how much better that model's fine-tuned system scores in ROUGE-L. Points above the line are pairs the ranking orders correctly. Filled markers are *resolved* pairs, whose ROUGE-L difference has a 95% bootstrap interval over the 500 articles that excludes zero. Adapted BPB gets 37 of 39 pairs right, including 21 of the 22 resolved pairs; HellaSwag gets 34; GSM8K gets 18 of 38, and all eleven of its pairs with gaps between 15 and 25 points fall below the line.

The fine-tuning task. We fine-tune fifteen of the seventeen models (Table 8) to write an article's opening paragraph, its lede, from the rest of the article. Gemma-4-31B and Qwen-3.5-35B-MoE are left out because they do not fit on one GPU. All fifteen use the same LoRA recipe and are scored by ROUGE-L on 500 held-out articles that are not part of the BPB test set. The analysis was written before the jobs ran (Appendix D). We chose this task because every article provides its own reference, the lede written by the journalist, so no annotation is needed, and writing a good lede requires reading the whole article and writing in the style of the domain. ROUGE-L, the longest common subsequence shared with the reference, is the standard automatic score for this kind of summarisation [27].

Statistics. We compare rankings mainly by *pairwise accuracy*, the fraction of model pairs that a ranking orders in the same way as a reference. Its 95% intervals come from a bootstrap over models rather than over pairs, because pairs that share a model are not independent (20,000 draws). Intervals for a single fine-tuned pair resample its 500 evaluation articles. Pairwise accuracy only counts *size-matched* pairs, meaning two models whose parameter counts differ by less than a factor of two. When one model is more than twice the size of the other, the larger one usually wins, and a ranking would get the pair right just by following size.

4 Results

4.1 Adapting to the domain predicts fine-tuning on the task

A team will fine-tune whichever model it chooses, so the practical question is whether the cheap measurement can predict how that fine-tuning will turn out. Measured with no task and no labels, adapted BPB orders 0.949 of the 39 size-matched pairs in the same way as the fine-tuned systems (95% interval $[0.817, 1.000]$). HellaSwag orders 0.872 of them, while GSM8K and MMLU-Pro are at chance level with 0.474 and 0.513 (Figure 3, Table 1). On the resolved pairs, which the fine-tuning results themselves separate, adapted BPB gets 21 of the 22 right. Over all fifteen models its ranking agrees with the final order at Spearman 0.99: every model is within one rank of its final position, and eleven are exactly in place. For comparison, unadapted BPB reaches 0.71, GSM8K 0.59, MMLU-Pro 0.71 and HellaSwag 0.97. Under this recipe adapted BPB also beats ranking by model size, by $+0.231$ in pairwise accuracy $[+0.059, +0.452]$. With a deliberately different

fine-tuning recipe, recipe B (four times the LoRA rank, three times the learning rate, half the batch size and a new seed), the order of the fine-tuned systems hardly changes (Spearman 0.982). Adapted BPB is still the best predictor at 0.923 and orders all 20 pairs that recipe B separates correctly.

Since the adaptation and the fine-tuning both use news, one might suspect that news is just predicting news. This is not the case. For the nine models that were both fine-tuned and adapted on all four corpora, BPB adapted on Reddit predicts the news fine-tuning as well as news BPB does (Spearman 0.983 each). Hacker News reaches 0.933, while the same corpora without adaptation reach only about 0.6. The prediction comes from adapting to fresh general text, not from adapting to the task's own domain. arXiv mathematics, which shifts the ranking towards reasoning (Section 4.3), predicts the news fine-tuning less well. A few minutes of unlabelled adaptation therefore capture most of what fine-tuning on the task reveals, while the reasoning benchmarks capture little of it.

Gemma-4-12B, ninth on GSM8K and eighth on MMLU-Pro among the fifteen models, produces the best fine-tuned system, ahead of Ministral-3-14B by $+0.018$ ROUGE-L $[+0.011, +0.025]$. The 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B makes no measurable difference after fine-tuning. Every model improved on its own base weights and beat the score of simply copying the article's first sentence (ROUGE-L 0.125).

	All 39 **pairs**	**95% CI**	22 **resolved pairs**	**Recipe B**
Adapted BPB	**0.949**	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Unadapted BPB	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 1: Which ranking predicts the fine-tuned systems? Fifteen models, each fine-tuned with the same recipe on the lede task and scored by ROUGE-L on 500 held-out articles. Each row gives the pairwise accuracy of one ranking over all size-matched pairs, with a bootstrap interval over models, and its accuracy on the 22 pairs whose fine-tuned difference is resolved by a bootstrap over articles. The last column scores the same rankings against systems trained with recipe B. GSM8K ties on one pair and is scored on 38.

4.2 Adaptation reorders the models

Adaptation lowers every model's loss, but it also changes their order (Figure 1A, Table 2). For the eleven models, the rankings before and after adaptation have a Spearman correlation of only 0.655 on news, 0.609 on Reddit and 0.682 on Hacker News, and one model moves seven places. The changes are uneven. On news, adaptation lowers loss by 3.6–5.4% for every Qwen, Llama and Ministral model, but by 17.0% for Gemma-4-31B, 24.5% for Gemma-4-12B and 36.4% for LFM2.5-1.2B. These three are the only models that move up, and every other model drops one or two places. Gemma-4-12B is ninth of eleven before adaptation and second after; before adaptation it ranks below two models with fewer than 2B parameters. LFM2.5-1.2B, the only non-Transformer, is last before adaptation at 1.34 bits per byte, behind a 0.5B Qwen at 0.93, and ahead of it after adaptation.

We attribute this to familiarity. Loss before adaptation mixes two things: whether a model already writes like the corpus, and whether it is a capable model. Figure 4 separates them by plotting loss against position inside an evaluation block. Each block is read with no earlier context, so the model starts every block cold. Before adaptation, Gemma-4-12B starts a block at 2.83 bits per byte, while Qwen-2.5-1.5B, eight times smaller, starts at 2.04. Gemma stays behind for roughly the first half of the block and is ahead by the end, 0.72 against 0.77. An average over the block therefore mostly measures how quickly a model gets used

Model	Params (B)	News BPB (rank)		Moves	Gain	Benchmark score, % (rank)		
		unadapted	adapted			HellaSwag	GSM8K	MMLU-Pro
Gemma-4-31B	31.4	0.752 (4)	0.624 (1)	↑**3**	**17.0%**	87.9 (1)	85.0 (3)	57.1 (2)
Gemma-4-12B	12.0	0.866 (9)	0.653 (2)	↑**7**	**24.5%**	84.0 (3)	64.8 (7)	44.8 (7)
Qwen-3.5-35B-MoE*	34.7	0.707 (1)	0.678 (3)	↓2	4.1%	85.6 (2)	86.6 (1)	61.4 (1)
Ministral-3-14B	14.0	0.718 (2)	0.679 (4)	↓2	5.4%	82.9 (4)	80.8 (6)	52.8 (4)
Qwen-3.5-9B	9.0	0.740 (3)	0.708 (5)	↓2	4.4%	81.6 (5)	85.8 (2)	55.6 (3)
Qwen-2.5-7B	7.7	0.758 (5)	0.717 (6)	↓1	5.4%	80.3 (6)	81.3 (4)	47.2 (6)
Qwen-3.5-4B	4.2	0.782 (6)	0.749 (7)	↓1	4.2%	77.2 (7)	81.0 (5)	52.4 (5)
Llama-3.2-1B	1.2	0.811 (7)	0.771 (8)	↓1	4.9%	65.8 (9)	6.4 (11)	11.2 (11)
Qwen-2.5-1.5B	1.6	0.833 (8)	0.797 (9)	↓1	4.3%	68.0 (8)	61.6 (8)	25.6 (8)
LFM2.5-1.2B	1.2	1.340 (11)	0.852 (10)	↑1	**36.4%**	60.9 (10)	54.1 (9)	16.4 (9)
Qwen-2.5-0.5B	0.5	0.935 (10)	0.901 (11)	↓1	3.6%	51.8 (11)	34.3 (10)	16.0 (10)

Table 2: The eleven models ranked by every measure. Sorted by adapted news BPB (lower is better). *Moves* is the number of places a model gains or loses with adaptation, and *Gain* is the reduction in loss. Three models gain far more than the rest (the two Gemmas and LFM2.5-1.2B), and they are the only ones that move up. After adaptation the order is almost the same as HellaSwag's, while GSM8K and MMLU-Pro differ: Llama-3.2-1B, eighth by BPB and ninth on HellaSwag, is last on both. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters.

to unfamiliar text, such as the formatting, document conventions and register of this corpus. That is a real property, but a model choice should not depend on it. A short adaptation, the same for every model, provides this familiarity. The loss reduction it brings is largest at the start of the block but remains after it, so what is left is closer to capability. Unadapted loss is still meaningful, but it measures something else, namely how similar the model already is to the target text. It penalises most the models whose pre-training text differs most from the corpus, which is also why the non-Transformer, with its own training mix, is affected most.

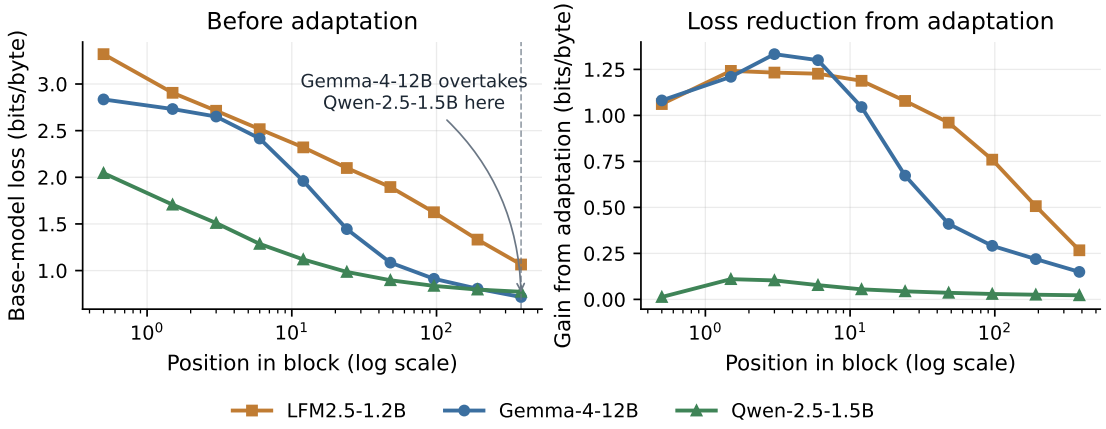


Figure 4: Loss before adaptation mostly measures how quickly a model adjusts to unfamiliar text. Each evaluation block starts with no earlier context. *Left:* loss before adaptation by position within a 512-token evaluation block, for Gemma-4-12B, Qwen-2.5-1.5B and LFM2.5-1.2B. Gemma-4-12B starts a block at 2.83 bits per byte, against 2.04 for Qwen-2.5-1.5B, and ends it at 0.72 against 0.77. A block average gives the start of the block a large weight, which is part of why Gemma-4-12B is ninth of eleven on unadapted BPB. *Right:* the loss reduction from adaptation at each position. It is largest at the start of the block but stays positive later, so adaptation teaches more than the corpus's opening conventions. This panel uses the three adapters from the earlier per-model search (Appendix C), so its gains differ from the fixed-recipe gains in Table 2; only the pattern across positions matters here.

4.3 The choice of text determines what the ranking measures

Of the three benchmarks, HellaSwag is the closest to language modelling. It asks which ending of a passage is most natural, and the harness scores each ending by its likelihood. Before adaptation, BPB on news, Reddit and Hacker News has a Spearman correlation of 0.63–0.72 with HellaSwag across the eleven models; after adaptation it is 0.96–0.98. On news, the adapted ranking orders 53 of the 55 model pairs the same way as HellaSwag, counting all pairs and not only size-matched ones (Table 2; the full matrix is Table 4 in Appendix B). Three independently collected corpora reach the same order once familiarity is removed, and it is the order of the benchmark closest to what BPB measures. Since both HellaSwag and BPB are based on likelihood, this agreement is expected. It shows that measuring on one's own fresh text loses nothing that HellaSwag captures, without the problems of a public benchmark.

The result depends on what the text is about. On general text, adaptation lowers the correlation with GSM8K on all three corpora and with MMLU-Pro on two of them. When the models are adapted on arXiv mathematics instead, BPB moves towards the reasoning benchmarks, with Spearman correlations of 0.91 with GSM8K and 0.93 with MMLU-Pro, up from 0.76 and 0.74, and ranking by mathematics BPB selects GSM8K's top model. With news the measurement agrees with the language benchmark, and with mathematics it agrees with the reasoning benchmarks. A benchmark cannot be redirected like this because its content is fixed, whereas the corpus is chosen by whoever is choosing the model.

4.4 Adapted BPB rankings are stable

A ranking is only useful if it is repeatable. We ran three preregistered seed replicates of all 17 news runs; the seed sets the adapter's initialisation and dropout, and the data splits and test set stay fixed. The median standard deviation of adapted BPB within a model is 0.000087, against a spread of 0.074340 between models, a ratio of 0.0012, and none of the 40 size-matched pairs changes order across seeds. The only model with visible seed noise is Gemma-4-12B, which varies by up to 1.24%, so gaps below 2% are best treated as ties. Re-running two models on different hardware reproduced unadapted BPB exactly and adapted BPB to within 0.0003, and repeating a run with a different software stack reproduced it to within 0.01%. The ranking also carries over across domains: adapted rankings agree at $\rho = 0.991$ between news and Reddit, 0.955 between news and Hacker News and 0.964 between Reddit and Hacker News (Table 6 and Figure 5, Appendix C). The unadapted ranking is just as stable across these corpora (0.982, 0.964 and 0.973), so adaptation does not make the ranking more stable. It produces a different stable ranking, the one explained in Section 4.2 and shown in Section 4.3 to agree with HellaSwag. Only mathematics differs from the other corpora, and only after adaptation: its agreement with the general corpora drops from 0.91–0.97 unadapted to 0.74–0.84 adapted, because adapting on mathematics brings out a different capability.

Benchmark scores are less stable. The same released weights move by -0.60 to $+0.69$ GSM8K points when only the GPU changes from A100 to H100, and one decoding default shipped with a checkpoint was worth 9.86 GSM8K and 14.30 MMLU-Pro points for Gemma-4-12B in our re-score (Appendix F). The reason lies in how the two are computed. Both run on hardware with the same floating-point nondeterminism, since GPU reductions are not associative and their results can change with batch size and kernel choice [26]. A generative benchmark decodes greedily, so a tiny change in two nearly tied logits can flip a token. The model then continues from its own changed output, and over a chain-of-thought answer the difference can grow until it flips a right-or-wrong verdict; each of GSM8K's 1,319 questions is worth 0.076 points. BPB is computed with teacher forcing: every prediction is conditioned on the true text, not on the model's earlier outputs, so a small perturbation changes one log-probability slightly and does not propagate. It is also averaged over about 1.3 million token predictions on news. Decoding settings, which changed Gemma-4-12B's GSM8K score by ten points, play no role because nothing is generated. A generative benchmark score depends on the model, the harness, the decoding configuration and the hardware; BPB on a fixed corpus after a fixed adaptation

depends mainly on the model.

5 Conclusion

Base models are usually chosen with public benchmarks, which have three problems. They are *expensive to run well*: a generative benchmark has to decode every answer, and its score changes with the hardware and the decoding settings, while the evaluation a team really needs, fine-tuning every candidate on its task, costs a training run per model. They *do not reflect deployment*, since they test a model as it ships while teams first adapt it to their own text. And they are *contaminated*: they are public, vendors optimise for them, and their questions can end up in pre-training data.

We propose a dynamic evaluation that addresses all three. Every candidate is adapted in the same way to fresh, decontaminated text from the target domain and ranked by its held-out loss in bits per byte. This is the loss each candidate is adapted to minimise, measured per byte so that models with different tokenizers can be compared.

It is cheap. Adaptation takes minutes to an hour per model on one GPU and needs no labels, yet it predicts the outcome of the expensive evaluation. All fifteen fine-tuned systems finish within one rank of the position adapted BPB gave them (Spearman 0.99), compared with 0.59–0.71 for the reasoning benchmarks, and adapting on Reddit instead of news predicts the news fine-tuning equally well. The full method tunes each model separately; because of limited compute, the runs here share one recipe, and a very different fine-tuning recipe leaves the order of the systems at Spearman 0.982.

It reflects deployment. Each model is measured after adaptation, on the kind of text it will be used on. Adaptation removes the advantage of already knowing the text's style and changes the ranking: Gemma-4-12B is ninth of eleven before adaptation, second after, and produces the best of the fifteen fine-tuned systems. The choice of text decides what the ranking measures. On general text it agrees with HellaSwag (Spearman 0.96–0.98), and on mathematics with GSM8K and MMLU-Pro (0.91 and 0.93).

It is clean. The text is dated after the models' release or screened for membership, and deduplicated, so no model has seen it. A new corpus can be collected for every domain and every evaluation.

It is stable. Because BPB is computed with teacher forcing, it is repeatable: seed-to-seed variation is about a thousandth of the spread between models. A different seed moves adapted BPB by up to 1.24% (Gemma-4-12B; the median model, 0.01%), so we treat a gap under 2% as a tie. The same weights, by contrast, move by up to 0.69 GSM8K points when only the hardware changes.

Our experiments cover seventeen base models, one fine-tuning task and one shared recipe per stage. Post-trained checkpoints can be evaluated in the same way. The released benchmark keeps one leaderboard per corpus and recipe: a news track that we run, and a mathematics track on arXiv titles and abstracts, whose corpus is public.

References

[1] Shannon, C. E. (1951). Prediction and entropy of printed English. Bell System Technical Journal, 30:50–64.

[2] Mielke, S. J. (2019). Can you compare perplexity across different segmentations? Technical note.

[3] Gao, L., et al. (2021). The Pile: An 800GB Dataset of Diverse Text for Language Modeling. arXiv:2101.00027.

[4] Ahia, O., et al. (2023). Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models. EMNLP. arXiv:2305.13707.

[5] Magnusson, I., et al. (2023). Paloma: A Benchmark for Evaluating Language Model Fit. arXiv:2312.10523.

[6] Carlini, N., et al. (2021). Extracting Training Data from Large Language Models. USENIX Security.

[7] Carlini, N., et al. (2022). Quantifying Memorization Across Neural Language Models. ICLR 2023. arXiv:2202.07646.

[8] Lee, K., et al. (2022). Deduplicating Training Data Makes Language Models Better. ACL. arXiv:2107.06499.

[9] Shi, W., et al. (2023). Detecting Pretraining Data from Large Language Models. ICLR 2024. arXiv:2310.16789.

[10] Zhang, J., et al. (2024). Min-K%++: Improved Baseline for Detecting Pre-Training Data from Large Language Models. ICLR 2025. arXiv:2404.02936.

[11] Xu, C., et al. (2024). Benchmark Data Contamination of Large Language Models: A Survey. arXiv:2406.04244.

[12] Rae, J. W., et al. (2021). Scaling Language Models: Methods, Analysis & Insights from Training Gopher. arXiv:2112.11446.

[13] Gao, L., Biderman, S., et al. (2024). A Framework for Few-Shot Language Model Evaluation (lm-evaluation-harness). EleutherAI.

[14] Heineman, D., Hofmann, V., Magnusson, I., et al. (2025). Signal and Noise: A Framework for Reducing Uncertainty in Language Model Evaluation. NeurIPS 2025.

[15] Thrush, T., Potts, C., and Hashimoto, T. (2024). Improving Pretraining Data Using Perplexity Correlations. arXiv:2409.05816.

[16] Zhang, G., Dominguez-Olmedo, R., and Hardt, M. (2025). Train-before-Test Harmonizes Language Model Rankings. arXiv:2507.05195.

[17] Ruan, Y., Maddison, C. J., and Hashimoto, T. (2024). Observational Scaling Laws and the Predictability of Language Model Performance. NeurIPS 2024.

[18] Gadre, S. Y., Smyrnis, G., Shankar, V., et al. (2025). Language Models Scale Reliably with Over-Training and on Downstream Tasks. ICLR 2025.

[19] Wang, Y., et al. (2024). MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. NeurIPS D&B. arXiv:2406.01574.

[20] Zellers, R., et al. (2019). HellaSwag: Can a Machine Really Finish Your Sentence? ACL. arXiv:1905.07830.

[21] Cobbe, K., et al. (2021). Training Verifiers to Solve Math Word Problems (GSM8K). arXiv:2110.14168.

[22] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2018). Dynamic Evaluation of Neural Sequence Models. ICML 2018. arXiv:1709.07432.

[23] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2019). Dynamic Evaluation of Transformer Language Models. arXiv:1904.08378.

[24] Gururangan, S., et al. (2020). Don't Stop Pretraining: Adapt Language Models to Domains and Tasks. ACL. arXiv:2004.10964.

[25] Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. KDD. arXiv:1907.10902.

[26] He, H., and Thinking Machines Lab (2025). Defeating Nondeterminism in LLM Inference. Thinking Machines Lab: Connectionism. https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/.

[27] Lin, C.-Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. ACL Workshop on Text Summarization Branches Out.

[28] Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022. arXiv:2106.09685.

[29] Dubey, A., et al. (2024). The Llama 3 Herd of Models. arXiv:2407.21783.

[30] Yang, A., et al. (2024). Qwen2.5 Technical Report. arXiv:2412.15115.

[31] Gemma Team (2026). Gemma 4 Technical Report. arXiv:2607.02770.

[32] Liu, A. H., et al. (2026). Ministral 3. arXiv:2601.08584.

[33] Qwen Team (2026). Qwen3.5: Towards Native Multimodal Agents. Official release and model card. https://qwen.ai/blog?id=qwen3.5.

A Related Work

From perplexity to comparable loss. Language modelling was grounded in byte- and character-level entropy from the start [1], but token-level perplexity depends on each model's vocabulary [2, 4]; normalising by UTF-8 bytes restores a common unit [3, 5], and loss-based metrics carry better signal and lower noise than accuracy across 465 models [14]. Our contribution is the operational protocol around the metric, not the metric.

Where the loss is measured matters. Thrush et al. turn the varying correlation between per-domain perplexity and benchmark performance into a pretraining-data selection rule [15]. Their estimand is the complement of ours: they fix the benchmark suite and choose data to *train* one strong model, where we fix the candidate pool and ask how the *selection ranking* moves.

Equal preparation. Dynamic evaluation updates a model on recent test history [22, 23] and domain-adaptive pretraining improves fit to a target distribution [24]. Closest to our motivation, *train-before-test* gives 61 models

identical supervised preparation across 24 tasks and raises cross-benchmark Kendall agreement from 0.52 to 0.76 [16]; they rank post-fine-tuning accuracy on public labelled benchmarks, where we rank held-out fit after *unsupervised* adaptation to a fresh corpus, across tokenizers. The core idea, prepare every candidate identically before comparing them, is theirs; what we add is doing it without labels, on text no vendor has seen, in a unit that compares tokenizers, and checking the result against real fine-tuned systems. Controlled families show a smooth loss-to-error relation [18, 17]; we test it *across* families.

Contamination. Models memorise duplicated pretraining text [6, 7], so deduplication and contamination analysis are necessary [8, 11]; reference-free membership signals are useful screens [9, 10] but prove nothing about a proprietary training set, so we use them only for conservative quarantine.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026[†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv `math.*`, 9 Jun–20 Aug 2026[‡]	3,207	3,207	–	15 MB	Post-release scrape + GSM8K shingle screen

Table 3: Corpora. All four receive identical exact-deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in *contamination* control, deliberately. The reference-free screen ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort's release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set, and they are protected by construction. Reddit and Hacker News comments are grouped to document length. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

B The Full Benchmark Alignment

		Rank correlation (ρ) with			Pairs ordered the same way, of 55		
Corpus	**Reading**	GSM8K	MMLU-Pro	HellaSwag	GSM8K	MMLU-Pro	HellaSwag
News	zero-shot	0.800	0.845	0.718	45	45	44
	adapted	0.700	0.773	0.982	42	44	53
Reddit	zero-shot	0.745	0.782	0.655	43	43	42
	adapted	0.655	0.745	0.964	41	43	52
Hacker News	zero-shot	0.800	0.809	0.627	44	44	43
	adapted	0.745	0.855	0.973	44	46	53
Math (arXiv)	zero-shot	0.764	0.736	0.482	43	43	40
	adapted	0.909	0.927	0.818	49	49	46

Table 4: Adaptation turns the ranking toward the benchmark that matches the text. Rank correlation of BPB with each benchmark over the eleven models, before (zero-shot) and after adaptation, on every corpus; ρ is signed so that 1 means lower BPB goes with a higher score. On the three general corpora adaptation lifts HellaSwag from 0.63–0.72 to 0.96–0.98 while GSM8K falls; on arXiv mathematics it lifts GSM8K and MMLU-Pro instead. The last three columns count, of all 55 pairs of the eleven (not only size-matched ones), those that BPB orders the same way as each benchmark: adapted news BPB agrees with HellaSwag on 53.

Table 4 reports all twenty-four (corpus $\times$ reading $\times$ benchmark) cells over the full 55 pairs; Table 5 prices each reading's top-1 pick against each benchmark's own. The rotation (Appendix C) is legible column by column: adapting on general text raises HellaSwag agreement and lowers the GSM8K column on every corpus and MMLU-Pro on two of the three general-text corpora; adapting on mathematics reverses the GSM8K half. The pair columns show how little of this a rank correlation conveys: across its eight rows the GSM8K column spans 41 to 49 of 55 pairs.

Corpus and reading	Picks	Accuracy given up (points)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 5: What choosing on BPB costs you on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that costs against each benchmark's own top-1 model. Only three distinct picks occur in eight rows, which is itself the finding: the adapted ranking selects Gemma-4-31B on every general-text corpus (the HellaSwag leader outright, at 1.6 GSM8K and 4.3 MMLU-Pro points), while adapting on mathematics recovers the GSM8K and MMLU-Pro leader at zero regret. The corpus, not the metric, decides which column you pay in.

C Robustness and Measurement Artefacts

Corpora compared	**Rank agreement** (Spearman)		**Value agreement** (Pearson)	
	unadapted	adapted	unadapted	adapted
News–Reddit	0.982	0.991	0.996	0.992
News–Hacker News	0.964	0.955	0.994	0.977
News–arXiv maths	0.909	0.755	0.964	0.839
Reddit–Hacker News	0.973	0.964	0.998	0.990
Reddit–arXiv maths	0.918	0.736	0.971	0.882
Hacker News–arXiv maths	0.973	0.836	0.982	0.930

Table 6: The ordering is a property of the models, not of one corpus. Agreement between the eleven models' BPB on two corpora. The three general-text corpora rank the models almost identically before and after adaptation. Mathematics is the telling row: adaptation makes it agree *less* with the general corpora, because adapted on mathematics the ranking follows a different capability (Section 4.3).

Stability across domains. All eleven models ran on three independently collected domains under the same recipe (Tables 3 and 9). Adaptation genuinely changes the choice: unadapted and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places, and the adapted ordering then holds across domains: news and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news against Hacker News gives 0.955 and Reddit against Hacker News 0.964, and the agreement is not merely ordinal (Figure 5). **So does the zero-shot ordering**: 0.982, 0.964 and 0.973 on the same three comparisons. The tempting story is that adaptation buys stability, and it does not; it buys a *different* stable ordering, and Section 4.2 is the evidence that it is the better of the two.

Why the search gives every model the same number of trials. The search of Phase 2 uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6}–10^{-3}, log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5}–10^{-1}, log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and promotes the best third at each rung; and the objective is BPB on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (`results/legacy_sweep/*/trials.csv`), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner, while the smaller

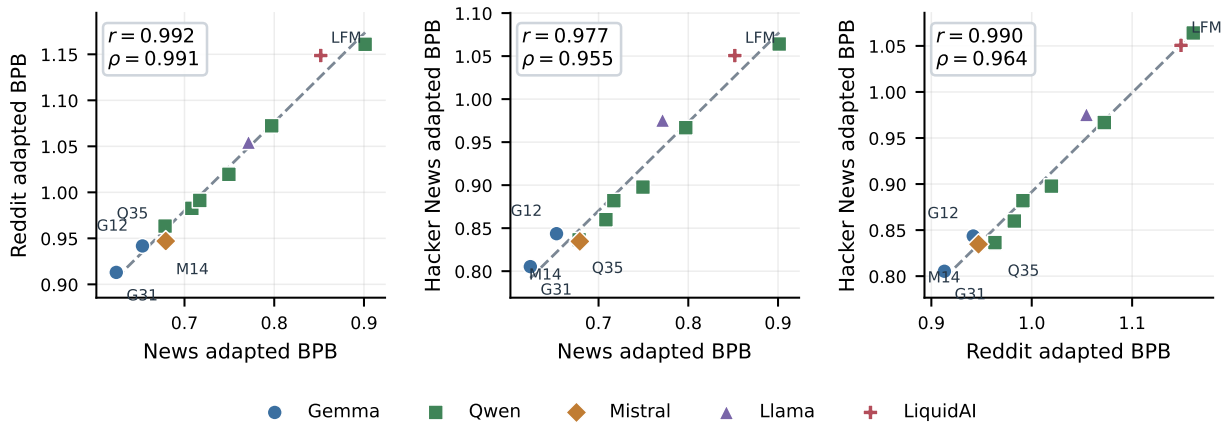


Figure 5: Adapted BPB across domains. Each point is one model, its adapted BPB on two corpora. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between the values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

models' winners came from much later in their searches. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. Apart from the right panel of Figure 4, which uses three of its adapters to show where in a block adaptation gains, none of those adapters produces a number in the paper; the reported results use the hand-picked configuration of Phase 2.

Membership screens. The news corpus additionally passes three reference-free membership screens per model: Min-K++ (normalised likelihood of the lowest-probability K=20% target tokens), context sensitivity (target NLL with and without sampled same-corpus prefixes), and, for flagged candidates only, preference over two word-swap/placeholder perturbations. The global union quarantined 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the unperturbed original is expected (fluency alone produces it), so we draw no memorisation conclusion from it (`results/contamination/`).

The cold start, seen a second way. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens reproduces Figure 4's effect non-positionally: both models improve with context and the larger improves faster, so any protocol that scores short contexts under-rates the models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, which is the constant the block-alignment bound below uses.

Token-aligned blocks: a real bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text: 5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at 0.08× the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the bound is looser and worth stating: on the worst adjacent pair, Qwen2.5-3B over Falcon3-7B, whose BPB gap is only 0.29%, the bound reaches 0.61× the gap. It still **overturns no rank**, but byte-aligned blocking would remove the question, and we recommend it for future work.

Seed variation is negligible. Every adaptation cell ran with the `torch` RNG unseeded, so a preregistered check re-ran all 17 news models three times (51 cells, everything else pinned). Median within-model SD of adapted BPB is 0.000087 against a between-model SD of 0.074340, a ratio of 0.0012, and *no* pair of models within 2× in size (0 of 40) reverses its ordering across seeds. The spread is not uniform: Gemma-4-12B is the one model where seed noise is visible at all (SD 0.00315, 5.3× the next-largest spread).

	Out of the box			After	Adapted BPB
Model	raw loss	GSM8K	MMLU-Pro	**fine-tuning**	**predicted**
Gemma-4-12B	13	9	8	**1**	2
Llama-3.2-1B	10	15	15	**11**	11
LFM2.5-1.2B	15	12	13	**14**	14

Table 7: How a model looks is not how it finishes. Rank among the fifteen fine-tuned models (1 is best). The benchmarks misjudge where these models end up, and raw loss misjudges Gemma-4-12B by twelve places: it is mid-table on both benchmarks and near the bottom on raw loss, then builds the best system of all. Llama-3.2-1B is last on both benchmarks and finishes eleventh. Adapted BPB, measured before any task fine-tuning, places every model within a single rank of where it finishes: all fifteen, and eleven exactly.

Steering: direction, not magnitude. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on contamination-screened arXiv mathematics instead raises GSM8K agreement to 0.909 and restores the GSM8K oracle pick at zero regret, while three general-text corpora tried as controls all leave GSM8K at or below the do-nothing baseline. That is the rotation, and it is why the body reports only its direction. Because the steering hypothesis predicts a *joint* movement, the better-powered statistic is a difference in differences: (GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus, which is $+0.242$ pooled. Resampling models gives a strictly positive difference in 93.7% of draws; the stricter publisher-family clustering the mechanism itself implies gives 80.9% with not one negative draw in 20,000, but no longer clears 0.05 on any tie convention. A leave-one-family-out jackknife shows why: dropping the two Gemma models takes the pooled difference from $+0.242$ to $+0.056$. The remedy is more *families*, not more models.

Three reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, two orders of magnitude below the differences compared. Because the extension models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through the extension pipeline; the analysis refuses to pool the cohorts unless it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170). And because benchmark rankings are themselves estimates, recomputing selection accuracy while discarding pairs whose benchmark gap is smaller than k combined standard errors leaves the result intact and slightly improves it: on GSM8K 0.909 at k=0 becomes 0.933 at k=3, and on HellaSwag 0.836 becomes 0.863, so the metric's errors concentrate in pairs the benchmark itself cannot separate.

D The Downstream Fine-Tune

Section 4.1 reports the result; this is how it was built, and what it does and does not license. It ran in two stages, each preregistered in code before its job returned: first the two pairs on which the selectors disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is chosen by anyone. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU *for this fine-tune* (its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation priced in Section 1) and are excluded by that rule, which costs one in-band pair of forty.

The task. Journalistic convention makes a news article's first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The BPB test blocks live in the *test* split, so the fine-tuning task is built from the validation split instead. That is verified rather than assumed: the test split reproduces from an independent reconstruction, and the adaptation pipeline's contamination pass (which would have split the corpus differently) is off by default and

recorded as off in every published cell. Evaluation documents are then hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four articles were dropped by that second check and none by the first.

Equal preparation. Both members of a pair get the paper's own adaptation recipe unchanged (LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch 32, 250 steps) with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. Nothing is tuned on either model, because a per-model search is exactly the cost this protocol exists to avoid. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model's realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint's own generation configuration otherwise untouched, following Appendix F. The prompt is assembled so that the instruction always survives truncation: the *body* is cut to fit, never the instruction after it, because right-truncating the assembled prompt would delete the instruction from the longest articles at a different article count for each tokenizer: an asymmetry between the two models being compared. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01; an interval excluding zero resolves the pair for whichever reading predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort: the fine-tuned ROUGE-L is the outcome and every selector is scored with the paper's own statistic: pairwise accuracy within 2×, model-level cluster bootstrap, 20,000 draws (Table 1). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between $+0.041$ and $+0.153$.

A second recipe. Both readers of an earlier draft raised the same objection: the fine-tune reused the selector's own recipe, so a model that adapts quickly under it could win both. Recipe B changes every setting the two shared (LoRA rank 64, α=128, learning rate 3×10^{-4}, effective batch 16, 400 steps, a new seed) on the same task, data, decoding and metric, preregistered before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted BPB remains the top selector at 0.923 $[0.750, 1.000]$, ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here ($+0.179$ $[-0.036, +0.441]$), so that margin should be read as recipe-specific (Table 1).

What copying buys. A news body restates its own lede, so reference overlap is partly available by retrieval. Two reference points bound it: the article's first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying (inspection confirms they largely do copy), while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result: scoring the news fine-tune with adapted BPB from each corpus, on the ten size-matched pairs of the nine fine-tuned models that have all four, gives 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted readings reach 0.500–0.600. Fresh general text carries the prediction, and mathematics, which aims the ranking at reasoning (Section 4.3), carries it least. Ten pairs support no interval, so this is a direction, not a result.

What this does not show. One task, one domain, two recipes, one seed each. On this task adapted BPB ordered the fine-tuned systems better than any other selector and GSM8K and MMLU-Pro ordered them at chance; its margin over HellaSwag does not resolve. It is not a claim that adapted BPB predicts downstream utility in general, and the paper does not make one.

E The Cohorts

Table 8 is the extended 17-model cohort behind the seed replicates and, less the two models too large for one GPU, the fine-tune: every model, its parameter count and its adapted news BPB. Table 9 then gives the original eleven across all three general corpora and both BPB endpoints, which is the population every other analysis in the paper uses. Table 9 is

Model	Params (B)	Adapted BPB	Cohort
Gemma-4-31B	31.4	0.624	original
Mistral-Nemo-12B	12.3	0.646	**added**
Gemma-4-12B	12.0	0.653	original
Qwen-2.5-14B	14.8	0.675	**added**
Qwen-3.5-35B-MoE*	34.7	0.678	original
Ministral-3-14B	14.0	0.679	original
Qwen-3.5-9B	9.0	0.708	original
Qwen-2.5-7B	7.7	0.717	original
Falcon3-10B	10.4	0.728	**added**
Qwen-3.5-4B	4.2	0.749	original
Qwen-2.5-3B	3.1	0.756	**added**
Falcon3-7B	7.5	0.758	**added**
Llama-3.2-1B	1.2	0.771	original
SmolLM2-1.7B	1.7	0.788	**added**
Qwen-2.5-1.5B	1.6	0.797	original
LFM2.5-1.2B	1.2	0.852	original
Qwen-2.5-0.5B	0.5	0.901	original

Table 8: The extended cohort, in full. All 17 models ordered by adapted news BPB. "added" marks the six models admitted for the size-matched extension on preregistered size-adjacency and publisher-spread criteria; the other eleven carry the four-corpus analyses. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

the full three-corpus leaderboard: both endpoints for all eleven models on news, Reddit and Hacker News, ordered by adapted news BPB. It is the artifact every eleven-model number in the paper derives from, as Table 8 is for the seventeen.

	News		Reddit		Hacker News	
Model	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LFM2.5-1.2B	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 9: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

Three checkpoints are excluded. The two OLMo-2 models fell under the preregistered infrastructure-fault rule: their adapted BPB cells exist and are unused, because the generation engine the study's evaluations share would not

initialise for them, identically before and after a cache rebuild. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window; it would have added about two size-matched pairs at the top of the size range. The OLMo exclusion removed a whole publisher, which is what the cohort most needs.

F The Benchmark Column Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test examples. All bf16, greedy, tokenizer-native BOS.

July's eleven-model column was scored on a host that no longer exists and generative scores do not pool across hosts: identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100, so it was re-scored in full: seventeen models $\times$ three benchmarks $=$ fifty-one cells, one host (A100-SXM-64GB), one build, with the design and the exclusion rule fixed in advance. Task definitions match exactly ($17/17$ identical task hashes on both hashed benchmarks) and every overlapping cell agrees with July to within 1.3 points, most to within 0.3.

Reaching that took one correction worth reporting, because the fault was in our own fix. Because the newer backend honours each checkpoint's shipped `generation_config.json` and the original did not, we replaced every checkpoint's generation settings with one identical policy, which also discarded `suppress_tokens`, a setting that is live for gemma-4-12B. That single omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points while leaving HellaSwag, scored by likelihood and never calling `generate()`, intact. Keeping the setting for all seventeen models reproduces July to $+1.21$ points on GSM8K, $+0.10$ on MMLU-Pro and -0.27 on HellaSwag. That GSM8K cell is the one outside the preregistered 1.0-point tolerance, within the allowance of one that the re-score declared in advance: the column is within its stated allowance rather than clean, and those are different claims. The lesson generalises: a wrapper imposed to remove a confound can be the confound, so the unmodified harness belongs in the design as a control.